Large-Stepsize GD on Multinomial Logistic Regression: Part II—On Distributed Optimization and Non-Separability

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Previously...

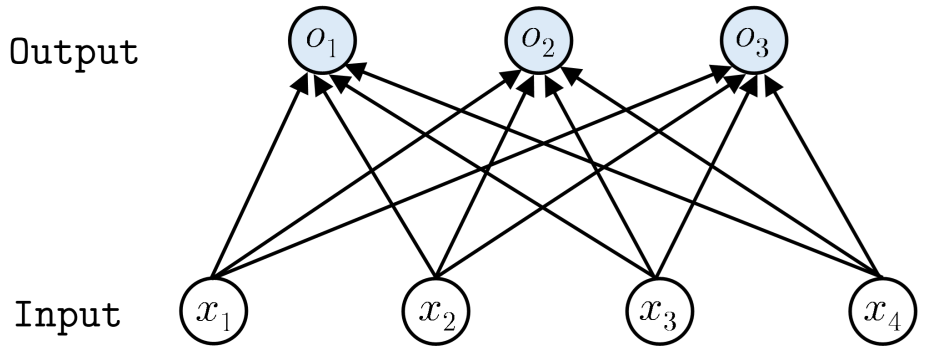
Multinomial Logistic Regression (Böhning, 1992)

Consider the K-class multinomial logistic regression problem:

$$\inf_{\mathbf{W}\in\mathbb{R}^{K\times d}} \mathcal{L}(\mathbf{W}) := \frac{1}{n}\sum_{i=1}^{n} \ell_i(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{n} \ell(\boldsymbol{\delta}^{(i)}, \widehat{\mathbf{y}}^{(i)}),$$

where

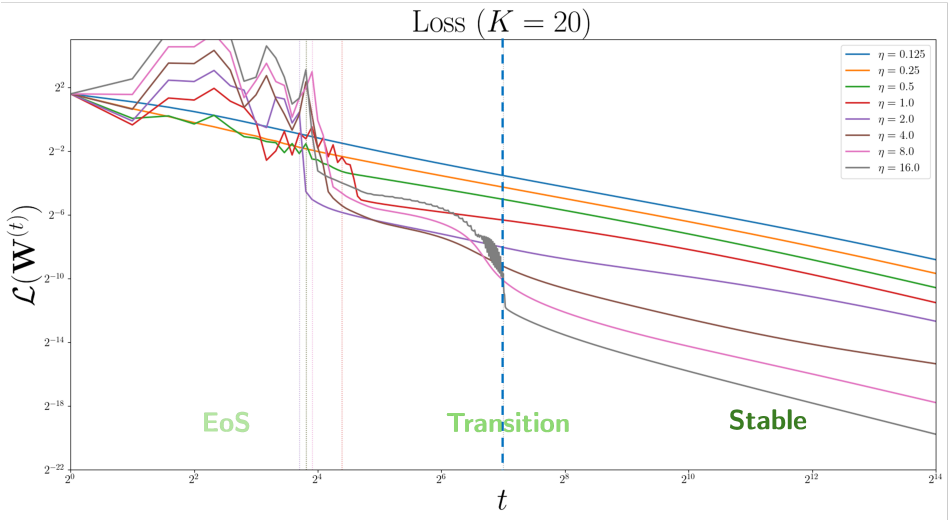
$$\ell(\boldsymbol{\delta}^{(i)}, \widehat{\mathbf{y}}^{(i)}) = -\sum_{k=1}^{K} \delta_k^{(i)} \log \widehat{y}_k^{(i)}, \quad \widehat{y}_k = \frac{\exp(\mathbf{w}_k^\top \mathbf{x})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x})}, k \in [K].$$

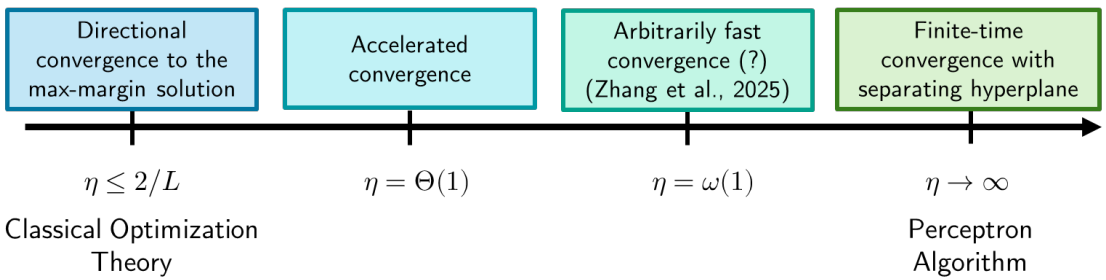


Previously...

Training process in two phases (Techniques from Wu et al. (2024)):

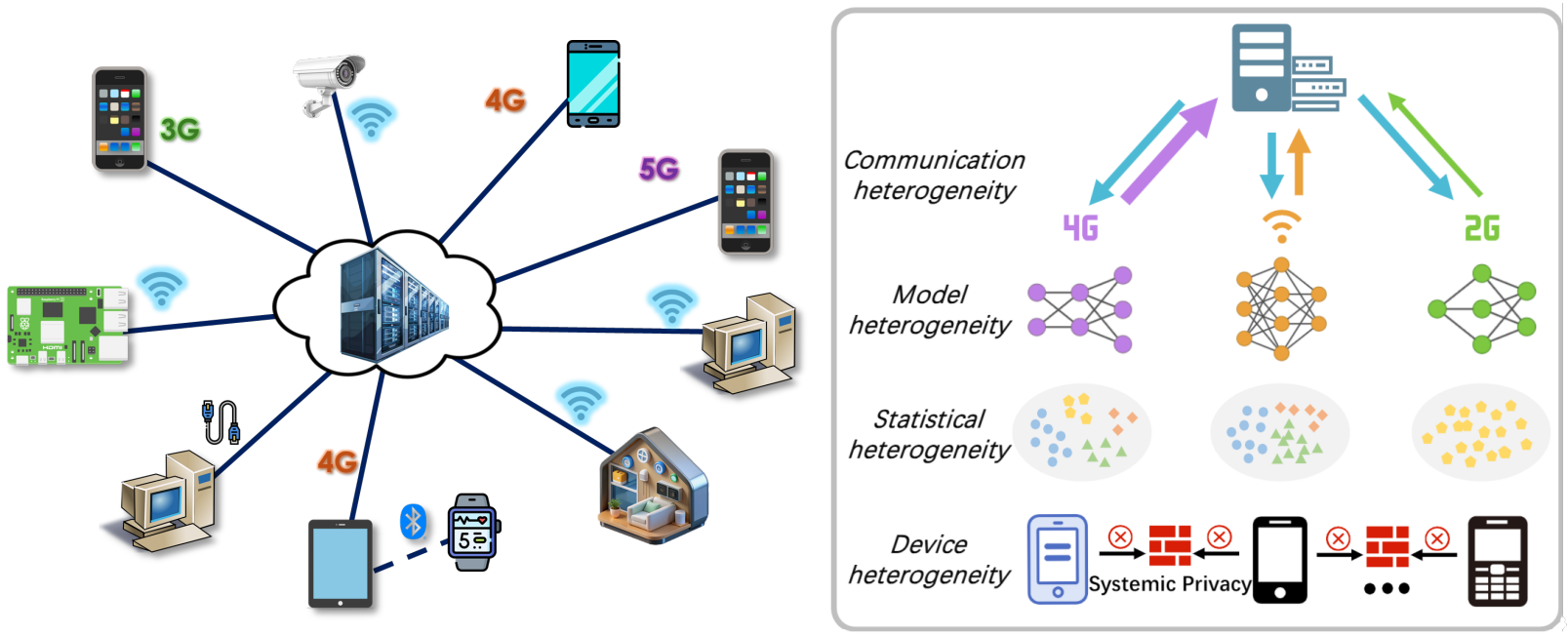
- **EoS phase**: non-monotonic loss with declining averaged loss.
- **Stable phase**: loss decreases monotonically, subsequently converge at a rate of $\mathcal{O}(1/t)$, very limited dependence on K, margin maximization.





Two Interesting Directions

1. **MLR with separable data under distributed optimization settings** (e.g. `FedAvg` by McMahan et al. (2017)).
   Previous work: Binary logistic regression with full homogeneity (Crawshaw et al., 2025).



Source: Ye et al. (2023).

Two Interesting Directions

2. **Dynamics of GD on non-separable datasets**.
   Previous work: Binary logistic regression under *non-separable* data (Meng et al., 2024).

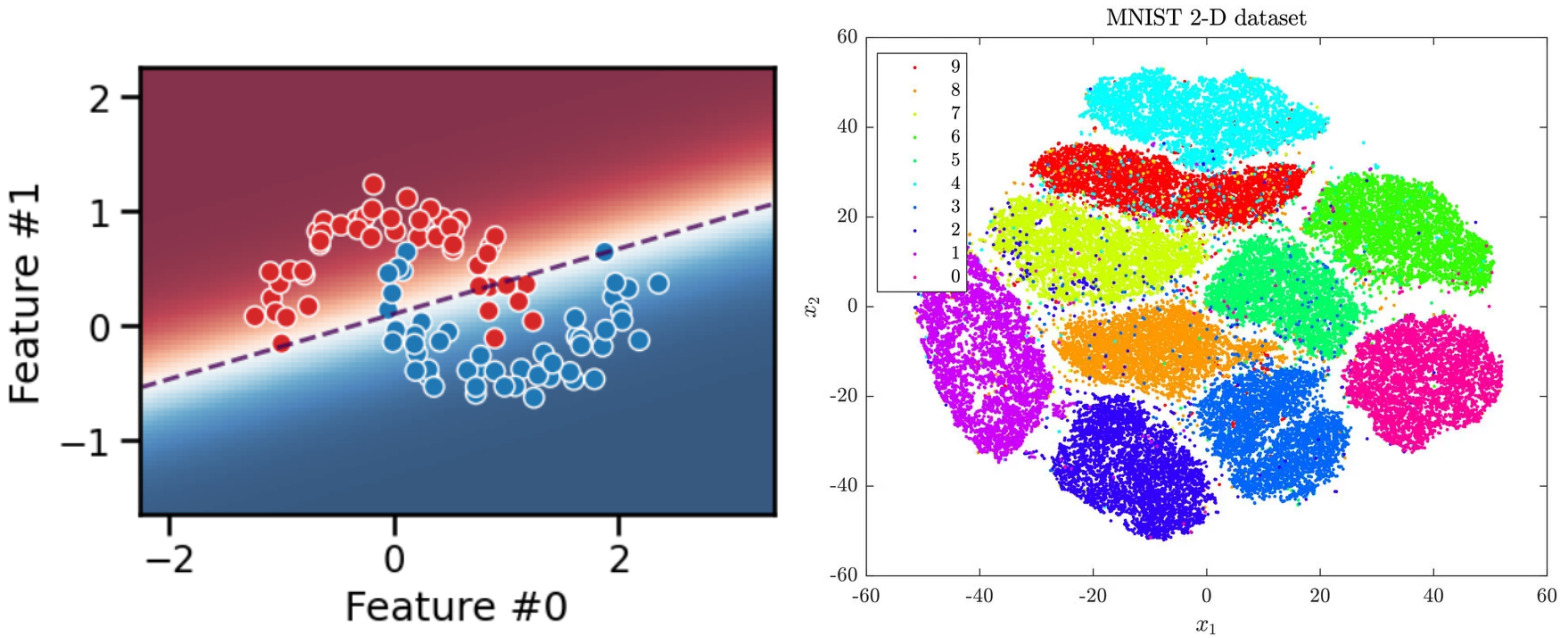


Figure: Example of a `TwoMoons` dataset and a 2D embedding of the `MNIST` dataset.

Contents

1. Large-Stepsize Federated Multinomial Logistic Regression Under Separable Data

2. Some Results of Multinominal Logistic Regression on Non-Separable Data
   - Convergence of GD under Adaptive Stepsize Schedule
   - Dynamics of GD under Non-Separability

3. Conclusion

Large-Stepsize Federated Multinomial Logistic Regression Under Separable Data

Rich Landscape Induced by Federated Learning

Challenges faced by federated learning:

- Decentralized data (access only to local dataset, for privacy);
- Communication budgets (network speed, bandwidth, etc.);
- Heterogeneity (hardware performance, iid/non-iid data distribution).

Research Question: Does the convergence of `FedAvg` with large stepsizes still hold under heterogeneous local updates and heterogeneous data distribution, and how does these heterogeneity effects influence the convergence bound?

Work in Submission:

H. F. Wong, H.-T. Wai, C.-Y. Yau, "Large Stepsizes Federated Learning on Logistic Regression with Linearly Separable Data: The Case of Heterogeneous Devices," The 65th IEEE Conference on Decision and Control (in submission), 2026.

Previous Works

- Controlling the client drift $\Rightarrow$ **Small stepsize regime**:
  1. `FedAvg` (McMahan et al., 2017): simple averaging of locally computed model updates, unstable and slow convergence under non-iid data;
  2. `FedProx` (Li et al., 2020): adds a proximal term to each client's local loss function to penalize large deviations from the global model, tackles device and data heterogeneity;
  3. `SCAFFOLD` (Karimireddy et al., 2020): applies variance control technique for estimating and correcting client drift, fastest convergence with consideration of data heterogeneity;
  4. `FedPD` (Zhang et al., 2021): reformulates FL as a primal-dual optimization problem, optimal rates, but can be complex to implement in practice.
- **Large-stepsize FL** – largely unexplored. Recent work by Crawshaw et al. (2025) focuses entirely on binary logistic regression with homogeneity, under identical local steps.

Our Work

- **Surprise!!** Plain FedAvg under MLR with large stepsizes is **inherently robust to heterogeneous local steps**, provided the data is well-conditioned (separable).
- We show that, when T_{avg} is large, or when the T_i's are similar to each other, `FedAvg` with large stepsizes still converges, and the **effect of heterogeneity vanishes asymptotically in the stable phase**;
- The average number of local updates T_{avg} dominates the convergence rate, while the **variation only affects the burn-in stage** (via $T_{\text{avg}}, T_{\max}, T_{\min}$ terms) and does not prevent convergence.

Problem Formulation

Centralized Objective

$$\inf_{\mathbf{W}\in\mathbb{R}^{K\times d}} \mathcal{L}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(\mathbf{W}),$$

where $\ell_i(\mathbf{W}) = -\log \frac{\exp(\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)})}{\sum_{k=1}^{K}\exp(\mathbf{w}_k^\top \mathbf{x}^{(i)})}$ is the cross-entropy loss.

Federated Setting (m devices)

Device i holds n_i samples $\{(\mathbf{x}^{(i_j)}, y^{(i_j)})\}_{j=1}^{n_i}$, with $\sum_i n_i = n$.

Local loss: $\mathcal{L}_i(\mathbf{W}) = \frac{1}{n_i}\sum_{j=1}^{n_i} \ell_{i_j}(\mathbf{W}).$

Global loss: $\mathcal{L}(\mathbf{W}) = \sum_{i=1}^{m} \frac{n_i}{n}\mathcal{L}_i(\mathbf{W}).$

Boundedness and Separability Assumptions

Assumptions on the Dataset

A1 (bounded feature norms). For any client $j \in [m]$, all local data samples $\{\mathbf{x}^{(i_j)}, y^{(i_j)}\}$, where $i \in [n_j]$, satisfy $\|\mathbf{x}^{(i_j)}\| \leq 1$.

A2 (global linear separability). There exists a margin $\gamma > 0$ and a global parameter matrix

$$\mathbf{W}_\star = [\mathbf{w}_{\star,1}^\top; \mathbf{w}_{\star,2}^\top; \ldots; \mathbf{w}_{\star,K}^\top]$$

with $\|\mathbf{W}_\star\|_F \leq 1$ such that for all clients $j \in [m]$ and all their local samples $(\mathbf{x}^{(i_j)}, y^{(i_j)}) \in \mathcal{D}_j$,

$$(\mathbf{w}_{\star,y^{(i_j)}} - \mathbf{w}_{\star,k})^\top \mathbf{x}^{(i_j)} \geq \gamma, \forall i = 1, \ldots, n \text{ and } k \neq y^{(i_j)}.$$

That is, the same model $\mathbf{W}_\star$ linearly separates the aggregated data from all m clients with margin γ.

Batch Gradient FedAvg for Distributed MLR

Require: Number of rounds R. Number of clients m.
Server input: initial $\mathbf{W}^{(0)}$ and global stepsize η_g.
Client i's ($i \in [m]$) input: local stepsize η_l.

for each round $r \leftarrow 1$ to R **do**
  communicate $\mathbf{W}^{(r-1)}$ to all clients $i \in [m]$
  on client $i \in [m]$ **in parallel do**
    initialize local model $\mathbf{D}_{i,0}^{(r)} \leftarrow \mathbf{W}^{(r-1)}$
    for each local step $\tau \leftarrow 1$ to T_i **do**
      compute the local gradient $\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) := \frac{1}{n_i} \sum_{j=1}^{n_i} \ell_{i_j}(\mathbf{D}_{i,\tau-1}^{(r)})$
      $\mathbf{D}_{i,\tau}^{(r)} \leftarrow \mathbf{D}_{i,\tau-1}^{(r)} - \eta_l \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)})$
    end for
    communicate $\Delta \mathbf{D}_i^{(r)} \leftarrow \mathbf{D}_{i,T}^{(r)} - \mathbf{W}^{(r-1)}$ and the local sample count N_i
  end on client
  $n \leftarrow \sum_{i=1}^{m} n_i$, $\mathbf{\Delta W}^{(r)} \leftarrow \frac{\eta_g}{n} \sum_{i=1}^{m} n_i \cdot \Delta \mathbf{D}_i^{(r)}$
  $\mathbf{W}^{(r)} \leftarrow \mathbf{W}^{(r-1)} + \mathbf{\Delta W}^{(r)}$
end for

Our Results

Theorem (Average Risk and Parameter Bound).

Suppose that Assumptions A1, A2 holds, for every $\eta_g, \eta_l > 0$, and $\bar{\eta} := \eta_g \eta_l$, we have

$$\frac{1}{R}\sum_{r=0}^{R-1} \mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{\|\mathbf{W}^{(0)}\|_F^2}{\bar{\eta} R T_{\min}} + \frac{1 + \ln^2(\gamma^2 \bar{\eta} R (K-1) T_{\text{avg}}) + \bar{\eta}^2 T_{\text{avg}}^2 + \eta_l^4 T_{\max}^4}{\gamma^2 \bar{\eta} R T_{\min}}.$$

Moreover, the final iterate satisfies that

$$\|\mathbf{W}^{(R)}\|_F \leq \|\mathbf{W}^{(0)}\|_F + \frac{\sqrt{2} + \ln(\gamma^2 \bar{\eta} R (K-1) T_{\text{avg}}) + \bar{\eta} T_{\text{avg}} + \eta_l^2 T_{\max}^2}{\gamma}.$$

Main Techniques

1. Define the **split reference** (Wu et al., 2024), $\mathbf{U} = \mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3$ with

$$\underbrace{\mathbf{U}_1 = \frac{\log((K-1)\gamma^2\bar{\eta}RT_{\text{avg}})}{\gamma}\mathbf{W}_\star}_{\text{large-margin part for small residual loss}}, \underbrace{\mathbf{U}_2 = \frac{\bar{\eta}T_{\text{avg}}}{\gamma}\mathbf{W}_\star}_{\text{total progression}}, \underbrace{\mathbf{U}_3 = \frac{\eta_l^2 T_{\max}^2}{\gamma}\mathbf{W}_\star}_{\text{drift absorption}}.$$

Using the server update

$$\boldsymbol{\Delta}\mathbf{W}^{(r)} = -\frac{\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \nabla\mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}).$$

We have the identity

$$\|\mathbf{W}^{(r+1)} - \mathbf{U}\|_F^2 - \|\mathbf{W}^{(r)} - \mathbf{U}\|_F^2 = \|\boldsymbol{\Delta}\mathbf{W}^{(r)}\|_F^2 + \underbrace{2\langle \boldsymbol{\Delta}\mathbf{W}^{(r)}, \mathbf{W}^{(r)} - \mathbf{U}_1\rangle}_{=:\mathcal{A}_1} + \underbrace{2\langle -\boldsymbol{\Delta}\mathbf{W}^{(r)}, \mathbf{U}_2 + \mathbf{U}_3\rangle}_{=:\mathcal{A}_2}.$$

Main Techniques

2. Bounding $\mathcal{A}_1$:

$$\mathcal{A}_1 = -\frac{2\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \langle \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}), \mathbf{W}^{(r)} - \mathbf{U}_1 \rangle$$
$$\leq \frac{2\bar{\eta}}{n}\sum_{i=1}^{m} n_i T_i \left(\mathcal{L}_i(\mathbf{U}_1) - \mathcal{L}_i(\mathbf{W}^{(r)}) \right) + \mathcal{E}^{(r)}$$

where

$$\mathcal{E}^{(r)} := \frac{\bar{\eta}}{2n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \|\mathbf{W}^{(r)} - \mathbf{D}_{i,\tau-1}^{(r)}\|_F^2.$$

The client drift bounding technique is inspired by Karimireddy et al. (2020),

$$\mathcal{E}^{(r)} \leq \frac{2\bar{\eta}\eta_l^2 T_{\max}^2}{n}\sum_{i=1}^{m}\sum_{t=1}^{T_i}\sum_{j=1}^{n_i}\left(1 - \widehat{y}_{i,j|\mathbf{D}_{i,t-1}^{(r)}}\right).$$

Main Techniques

3. Bounding $\mathcal{A}_2$:

$$\mathcal{A}_2 = \frac{2\bar{\eta}}{n}\left(\frac{\bar{\eta}T_{\text{avg}}}{\gamma} + \frac{\eta_l^2 T_{\max}^2}{\gamma}\right)\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \langle \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}), \mathbf{W}_\star \rangle$$
$$\leq \frac{2\bar{\eta}}{n}\left(\bar{\eta}T_{\text{avg}} + \eta_l^2 T_{\max}^2\right)\sum_{i=1}^{m}\sum_{\tau=1}^{T_i}\sum_{j=1}^{n_i}\left(\widehat{y}_{i,j|\mathbf{D}_{i,\tau-1}^{(r)}} - 1\right).$$

4. $\|\boldsymbol{\Delta}\mathbf{W}^{(r)}\|_F^2$ can be bounded as

$$\|\boldsymbol{\Delta}\mathbf{W}^{(r)}\|_F^2 \leq \left(\frac{\bar{\eta}^2}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i\right)\sum_{i=1}^{m}\sum_{\tau=1}^{T_i}\frac{n_i}{n}\|\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)})\|_F^2$$
$$\leq \frac{2\bar{\eta}^2 T_{\text{avg}}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i}\sum_{j=1}^{n_i}\left(1 - \widehat{y}_{i,j|\mathbf{D}_{i,\tau-1}^{(r)}}\right).$$



Our Results (cont.)

Theorem (Stable Phase).

Under Assumptions A1 and A2, for every $\eta_g, \eta_l > 0$, $\bar{\eta} = \eta_g \eta_l$, and $i \in [m]$, suppose that there exists $s \geq 0$ such that

$$\mathcal{L}(\mathbf{W}^{(s)}) \leq \min \left\{ \frac{\gamma T_{\text{avg}}}{40 n \eta_l T_{\max}^2 \sqrt{K}}, \frac{1}{2n}, \frac{1}{4 \bar{\eta} T_{\text{avg}}} \right\},$$

and the heterogeneity among clients satisfies $\max_i |T_i / T_{\text{avg}} - 1| \leq \gamma / (20\sqrt{K})$, then $(\mathcal{L}(\mathbf{W}^{(r)}))_{r \geq s}$ decreases monotonically.

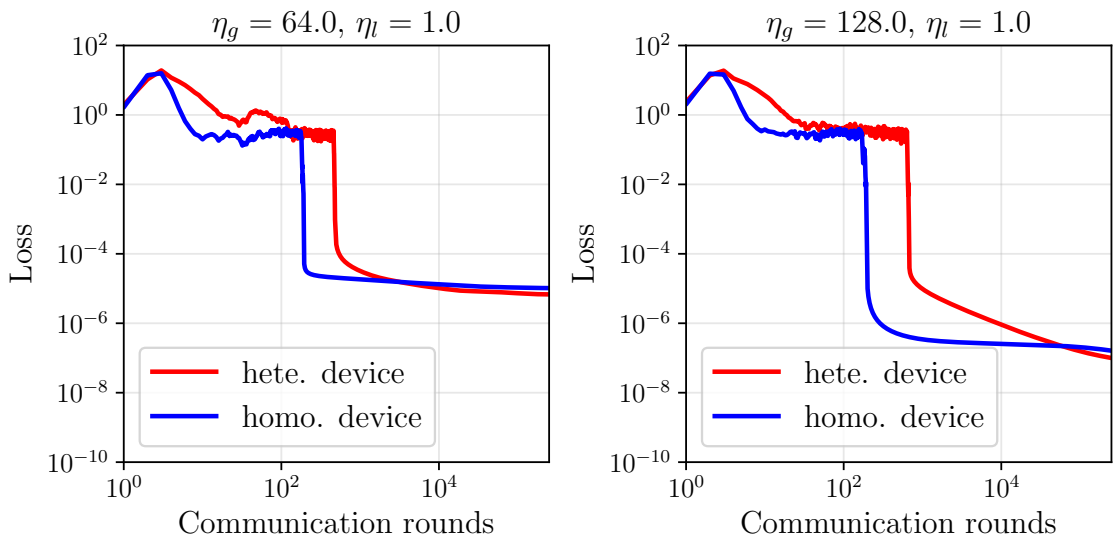
Theorem (Stable Descent).

Under the conditions in the previous Theorem, for every $r \geq s$ we have

$$\mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{16}{\bar{\eta} T_{\text{avg}} \gamma^2 (r - s)}.$$

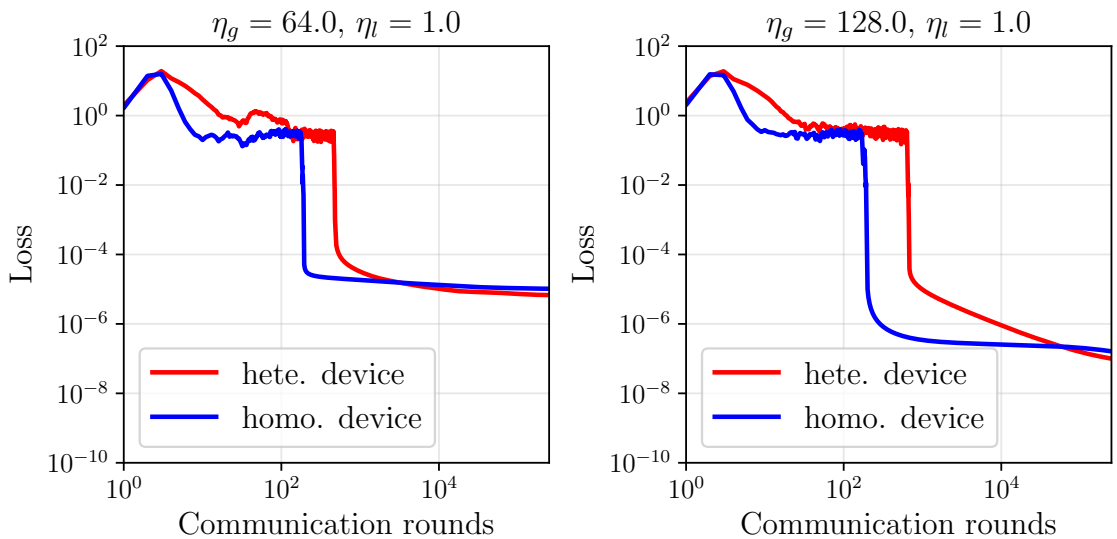
Effect of Heterogeneous Local Epochs: Setup

- Fix local stepsize $\eta_l = 1$ and global stepsizes $\eta_g \in \{64, 128\}$
- Communication rounds $R = 200,000$.
- (i) ***Heterogeneous*** local steps, where clients 1-4 perform $T_i = 4$ local updates per communication round, while clients 5-8 perform $T_i = 50$ updates, yielding $T_{\text{avg}} = 27$;
  (ii) ***Homogeneous*** local steps, where all clients perform $T_i = 27$ local steps per round.
- Dataset is partitioned among $m = 8$ clients; each client receives exactly 625 samples and at most with 2 classes for heterogeneity.



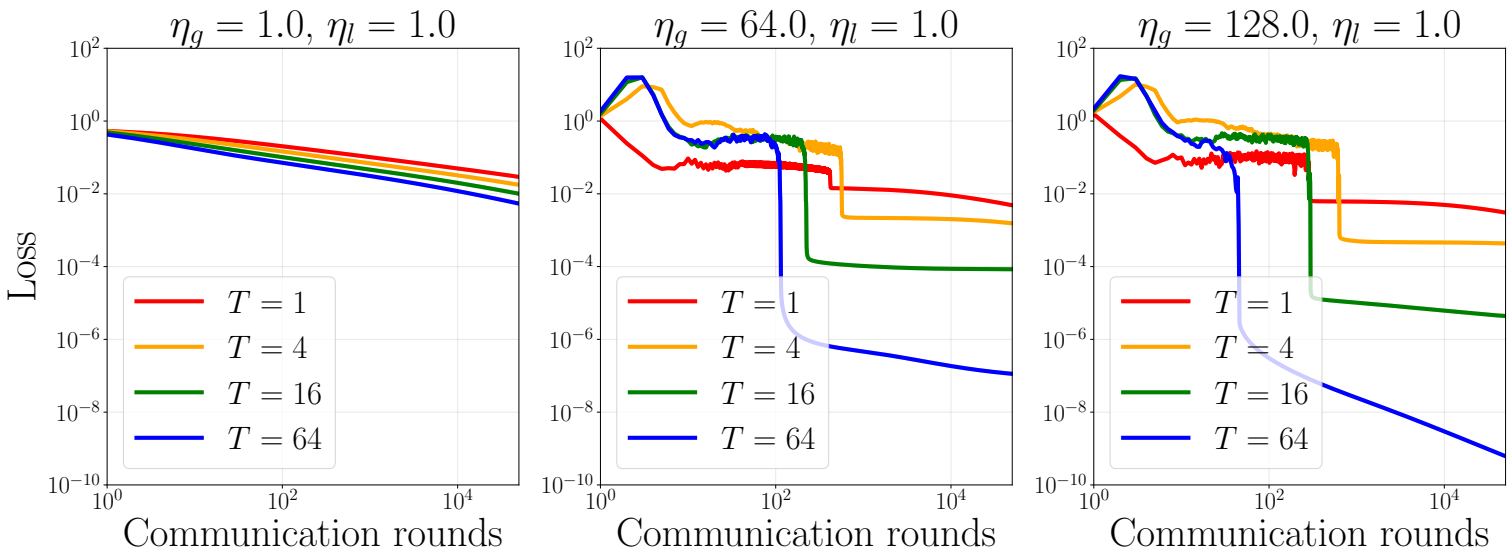
Effect of Heterogeneous Local Epochs: Setup

- Heterogeneous device: longer transient with larger fluctuations initially;
- After the transient phase as $R \gg 1$, the convergence behavior of both heterogeneous and homogeneous cases become indistinguishable – the asymptotic rate is dominated by $\mathcal{O}(1/(T_{\text{avg}}R))$, independent of the device heterogeneity.



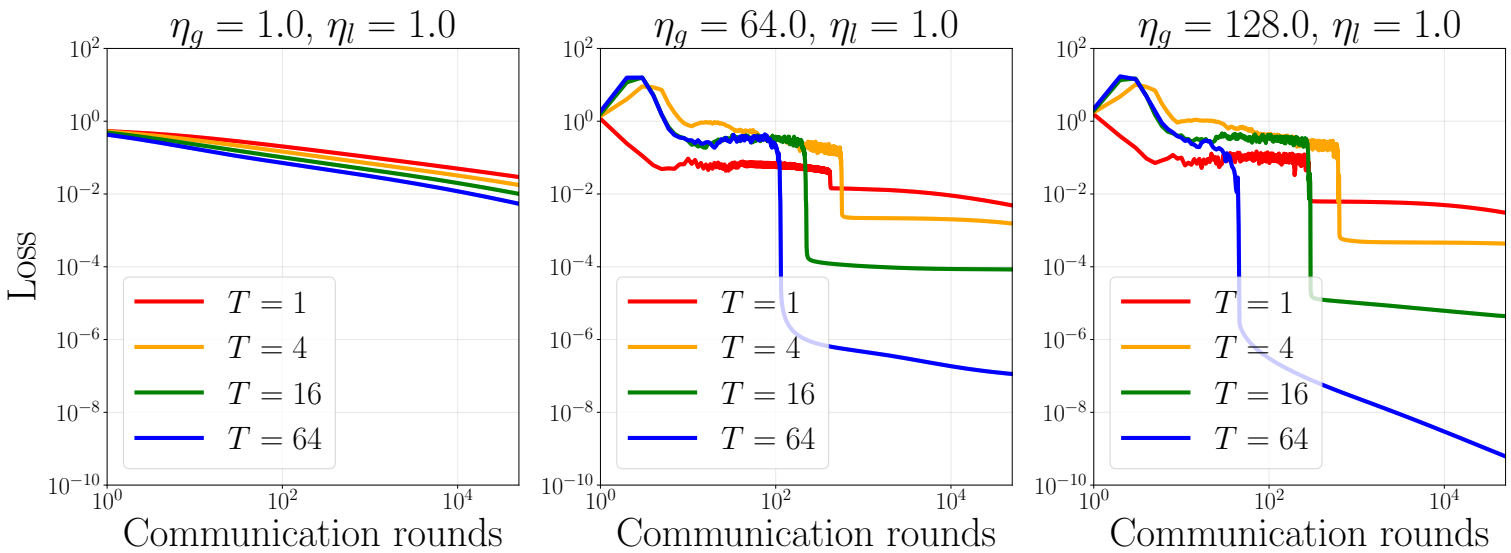
Effect of Large Local Steps under Label Imbalance: Setup

- Fix local stepsize $\eta_l = 1$ and global stepsizes $\eta_g \in \{1, 64, 128\}$
- Communication rounds $R = 50,000$.
- ***Homogeneous*** local steps, where all clients perform $T_i \in \{1, 4, 16, 64\}$ local steps per round.
- Dataset is partitioned among $m = 8$ clients; each client receives exactly 625 samples and at most with 2 classes for heterogeneity.



Effect of Large Local Steps under Label Imbalance: Setup

- Increasing T leads to a faster convergence.
- Benefits of using large stepsizes: $\eta_g \in \{64, 128\}$ significantly accelerate the convergence compared to the baseline case with $\eta_g = 1$.



Some Results of Multinominal Logistic Regression on Non-Separable Data

Previous Works & Our Attempt

Binary Case (Meng et al., 2024):

- **Separable Case**: GD with any constant stepsize converges (Wu et al., 2024). Adaptive schedule $\eta_t = \eta/\mathcal{L}(\mathbf{w}^{(t)})$ accelerates convergence (Zhang et al., 2025).
- **Non-Separable Case**: Cycles can emerge even below $2/\lambda$ for problems with feature dimension $d \geq 2$.

Research Question: Can we characterize the convergence behavior of gradient descent with large stepsizes on **non-separable multiclass problems**, and construct datasets that exhibit cyclic behavior with explicit dependence on the number of classes K?

Two Regimes:

- **Adaptive stepsize** – under what stepsize schedule does GD guarantee convergence?
- **Constant stepsize** – are cycles possible for $K \geq 3$?

Assumptions and Fundamental Properties

Assumption B1 (Bounded Feature Norms)

For all samples $(\mathbf{x}^{(i)}, y^{(i)})$, where $i \in [n]$, $\|\mathbf{x}^{(i)}\| \leq 1$.

Assumption B2 (Strict Non-Separability)

There is no $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top; \ldots; \mathbf{w}_K^\top] \in \mathbb{R}^{K \times d}$ such that for all i, $\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)} \geq \max_{k \neq y^{(i)}} \mathbf{w}_k^\top \mathbf{x}^{(i)}$.

Assumptions and Fundamental Properties

Lemma (Consequences of Non-Separability)

Consider the multinomial logistic loss $\mathcal{L}$ on the non-separable dataset satisfying Assumption B2. Let $S = \{\mathbf{W} \in \mathbb{R}^{d\times K} : \sum_k \mathbf{w}_k = \mathbf{0}\}$ and $V = \text{span}\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subseteq \mathbb{R}^d$. Then, the minimizer $\mathbf{W}_\star$ is finite on the subspace $S \cap V$.

Lemma (Weight Sum Preservation across Gradient Steps)

For the multinomial logistic loss $\mathcal{L}$, its gradient satisfies $\sum_{k=1}^K \nabla_{\mathbf{w}_k} \mathcal{L}(\mathbf{W}) = \mathbf{0}$. Consequently, the sum of the weight vectors is invariant under gradient descent, i.e.

$$\sum_{k=1}^K \mathbf{w}_k^{(t+1)} = \sum_{k=1}^K \mathbf{w}_k^{(t)}$$

for all $t \geq 0$.

Remarks. Unused data dimensions and translation invariance give rise to degrees of freedom. These lemmas are aimed at reducing the effective dimensionality.

Monotonic Convergence under Adaptive Stepsizes

Theorem (Stable Gradient Descent with Adaptive Stepsizes)

Consider the GD progression on the multinomial logistic loss with non-separable dataset satisfying Assumptions B1, B2 with

$$\mathbf{W}^{(t+1)} \leftarrow \mathbf{W}^{(t)} - \eta_t \nabla \mathcal{L}(\mathbf{W}^{(t)}).$$

For any initial point $\mathbf{W}^{(0)}$, and using the *adaptive* stepsize schedule $\{\eta_t\}_{t\geq 0}$, where $\eta_t = \gamma/(K\mathcal{L}(\mathbf{W}^{(t)}))$, and $\gamma \in (0, 1]$, then the loss $\mathcal{L}(\mathbf{W}^{(t)})$ decreases monotonically, and $\mathbf{W}^{(t)}$ converges to a finite global minimizer at a rate of $\mathcal{O}(1/t)$.

Discussions.

- Smaller loss allows the use of a larger stepsize, does not require prior knowledge of the global smoothness constant.
- Compared with Zhang et al. (2025), we operate on strictly **non-separable data**, **last iterate convergence** with rate $\mathcal{O}(1/t)$ and **finite minimizer**.

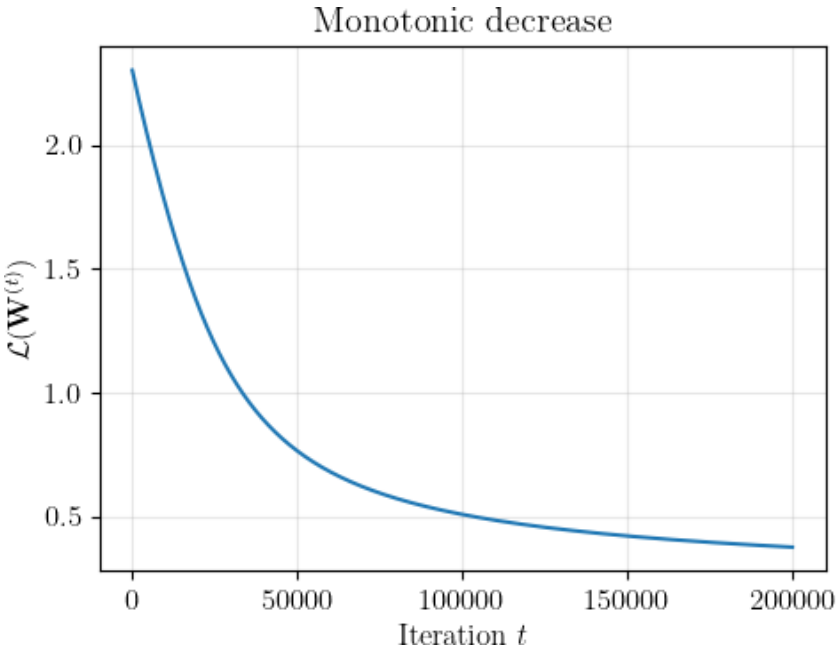
Proof Sketch

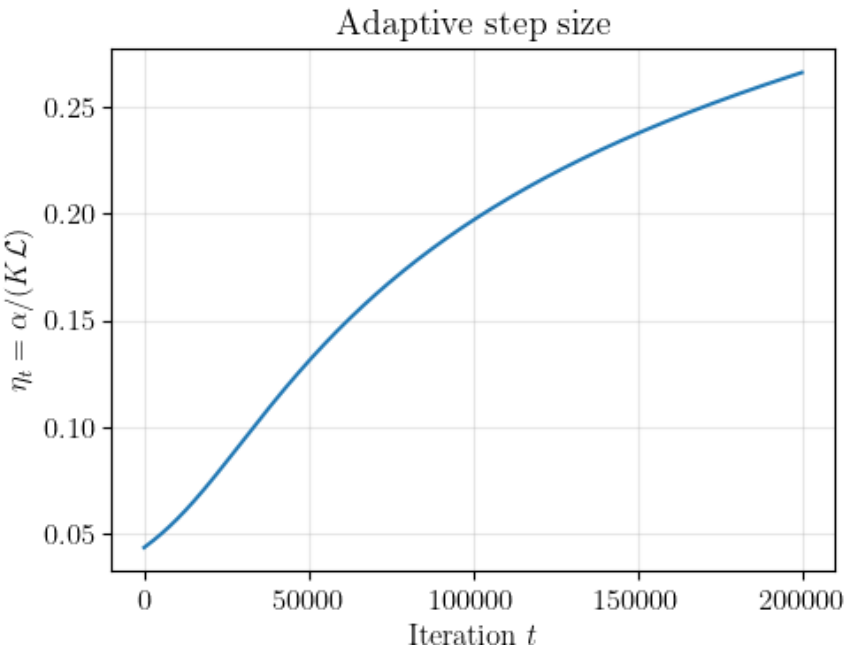
Three steps:

1. **Monotonic decrease.**
   - Taylor expansion + Hessian bound $\|\nabla^2 \mathcal{L}\|_{\text{op}} \leq 2\mathcal{L}$.
   - Assume $\mathcal{L}(\mathbf{W}^{(t+1)}) > \mathcal{L}(\mathbf{W}^{(t)})$.
   - With $\eta_t = \gamma/(K\mathcal{L}(\mathbf{W}^{(t)}))$ $(0 < \gamma \leq 1)$, algebra leads to a contradiction.
   - Hence $\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)})$.
2. **Gradient convergence.**
   - From descent: $\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{W}^{(t+1)}) \geq \frac{\beta \|\nabla \mathcal{L}\|_F^2}{\mathcal{L}(\mathbf{W}^{(t)})}$, $\beta > 0$.
   - Summing gives $\sum \|\nabla \mathcal{L}\|_F^2 < \infty$, so $\|\nabla \mathcal{L}\|_F \to 0$.
3. **Convergence to minimizer.**
   - Project onto $V \cap S$ (remove translation invariance + data-orthogonal components).
   - Coercivity $\Rightarrow$ iterates bounded.
   - Accumulation point has $\nabla \mathcal{L} = 0 \Rightarrow$ global minimizer (convexity).
   - Invariant sum $\Rightarrow$ full sequence converges.

Key tools: Hessian bound, coercivity, invariant sum, convexity.

Numerical Experiment on MNIST





Experimental Setup

- Dataset: MNIST (random subset of $5,000$ training samples and features are flattened to 784-dimensional vectors.
- Preprocessing: Each sample is rescaled with Euclidean norm ≤ 1.
- Adaptive stepsize schedule: $\eta_t = \gamma/(10 \cdot \mathcal{L}(\mathbf{W}^{(t)})), \gamma = 1$.
- 200,000 iterations.

Large Stepsize GD

Gradient Descent

$$\mathbf{w}^{(k)} \leftarrow \mathbf{w}^{(k-1)} - \eta \nabla \mathcal{L}(\mathbf{w}^{(k-1)}), k = 1, 2, \ldots, T$$

If $\mathcal{L}$ is L-Lipschitz smooth, then

- If $\eta < 2/L$, then the loss value decreases monotonically.
- If $\eta < 1/L$, then the loss will eventually converge to a local minimum.

Fact (Optimizing a Quadratic Cost Function)

Consider $f(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \mathbf{P}\boldsymbol{\theta} + \mathbf{q}^\top \boldsymbol{\theta} + r$, GD will diverge if any eigenvalue of $\mathbf{P}$ exceeds $2/\eta$.

Insights (Ahn et al., 2022; Meng et al., 2024)

Under certain assumptions, a necessary condition for $\mathbf{w}^*$ to be locally stable is to have a stepsize smaller than $2/\lambda$, where λ is the largest eigenvalue of the Hessian at $\mathbf{w}^*$.

Stability Analysis as a Dynamical System

Define

$$\mathcal{T}(\mathbf{w}) := \mathbf{w} - \eta \nabla \mathcal{L}(\mathbf{w}),$$

and

$$\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta \nabla \mathcal{L}(\mathbf{w}_t).$$

The fixed point $\mathbf{w}_\star$ satisfies

$$\mathcal{T}(\mathbf{w}) \approx \mathcal{T}(\mathbf{w}_\star) + \mathbf{J}_{\mathcal{T}}(\mathbf{w}_\star)(\mathbf{w} - \mathbf{w}_\star),$$

where $\mathbf{J}_{\mathcal{T}}(\mathbf{w}_\star) = \mathbf{I} - \eta \nabla^2 \mathcal{L}(\mathbf{w}_\star)$ is the Jacobian matrix of $\mathcal{T}$ at $\mathbf{w}_\star$.
Then, the linearized system is

$$\boldsymbol{\Delta}_{t+1} \approx \mathbf{J}_{\mathcal{T}}(\mathbf{w}_\star)\boldsymbol{\Delta}_t,$$

by letting $\boldsymbol{\Delta}_t = \mathbf{w}_t - \mathbf{w}_\star$.
Sufficient and necessary condition: modulus of all eigenvalues of the Jacobian matrix is less than 1. Hence, the stability requirement is given by $\eta < 2/\lambda_{\max}(\nabla^2 \mathcal{L}(\mathbf{w}_\star))$.

Constant Stepsize: Previous Works & Our Attempt

Binary Case (Meng et al., 2024): Cycles can emerge even below $2/\lambda$ for problems with feature dimension $d \geq 2$.

Table: Status of cycle existence for non-separable multiclass logistic regression with constant stepsize $\eta = \gamma/\lambda$ (where λ is the largest Hessian eigenvalue at the minimizer).

K	d	γ	Status	Reference
2	1	$(0, 1]$	✓ Proved	Theorem 4.8 + Meng et al. (2024)
2	1	$(1, 2]$	✓ Proved	Theorem 4.9 + Meng et al. (2024)
2	2	$(0, 1]$	✓ Proved	Theorem 4.10 + Meng et al. (2024)
2	≥ 2	$(0, 1]$	✓ Proved	Theorem 4.12 + Meng et al. (2024)
≥ 3	≥ 1	any	× Open	Hypothesis 4.13
≥ 3	$= K$	$(1, 2]$	✓ Proved	Theorem 4.14 (Our work)
≥ 3	$\geq K$	$(1, 2]$	✓ Proved	Lemma 4.11 + Theorem 4.14 (Our work)

A Weak Non-Separability Assumption

Assumption B3 (Weak Non-Separability)

There is no $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top] \in \mathbb{R}^{2\times d}$ such that for all i, $\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)} > \max_{k\neq y^{(i)}} \mathbf{w}_k^\top \mathbf{x}^{(i)}$.

Note: The analogous condition of B2, for $K = 2$ classes, is stronger than B3. B2 rules out even the possibility of a "weak" classifier that never strictly misclassifies any point.

Equivalence on 2-class MLR and Binary Logistic Loss

The following shows how a 2-class problem reduces to a 1-parameter problem in ϕ.

Lemma (Equivalence of 2-class Multinomial Logistic Loss and Binary Logistic Loss)

Let $\mathcal{L}(\mathbf{w}_1, \mathbf{w}_2)$ be the 2-class multinomial logistic loss and $\tilde{\mathcal{L}}(\phi)$ be the binary logistic loss with $\phi = \mathbf{w}_1 - \mathbf{w}_2$ and relabeled labels $\tilde{y}^{(i)} \in \{0, 1\}$. Then, under gradient descent with stepsize η on L, the difference $\phi^{(t)} = \mathbf{w}_1^{(t)} - \mathbf{w}_2^{(t)}$ evolves as gradient descent on $\tilde{\mathcal{L}}$ with stepsize 2η. The Hessian of $\mathcal{L}$ at $\mathbf{W}$ is with eigenvalues 0 and $2\lambda_\phi$, where λ_ϕ are the eigenvalues of $\nabla^2 \tilde{\mathcal{L}}(\phi)$.

Consequence. The 2-class MLR problem on $\mathbf{w}_1, \mathbf{w}_2$ is equivalent to binary logistic regression on ϕ with stepsize 2η.

Theorem (Stable Convergence of 2-class Multinomial Classification in 1D)

Consider GD on the 2-class multinomial classification task in 1D with non-separable data. Then, the GD iterates $\{(w_1^{(t)}, w_2^{(t)})\}_{t\geq 0}$ under $\eta \leq 1/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) converge to the minimizer $\mathbf{w}_\star = (\frac{s^{(0)}+\phi_\star}{2}, \frac{s^{(0)}-\phi_\star}{2})$ for any initialization $w_1^{(0)}, w_2^{(0)} \in \mathbb{R}$ and $s^{(0)} = w_1^{(0)} + w_2^{(0)}$, where $\phi_\star$ is given by the minimizer of the corresponding binary logistic loss $\tilde{\mathcal{L}}(\phi)$ on the equivalent non-separable dataset.

Proof Sketch. The 2-class problem reduces to binary logistic regression on $\phi = w_1 - w_2$ with stepsize 2η. For $\eta \leq 1/\lambda$, we have $2\eta \leq 2/\lambda = 1/\mathcal{L}''_\phi(\phi_\star)$. Theorem 1 of Meng et al. (2024) guarantees that $\phi^{(t)} \to \phi_\star$ for any initialization. The results follows from $s^{(t)} = w_1^{(0)} + w_2^{(0)}$ and $w_1 = (s+\phi)/2$, $w_2 = (s-\phi)/2$. □

Theorem (Existence of Cyclic Trajectory of 2-class Multinomial Logistic Regression in 1D)

For any $\gamma \in (1, 2]$, there exists a 1-dimensional non-separable 2-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size $\eta = \gamma/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period $k > 1$.

Proof Sketch. For any $\gamma \in (1, 2]$, Theorem 2 of Meng et al. (2024) constructs a 1D binary logistic regression dataset and an initialization such that GD with stepsize $\eta_\phi = \gamma/\mathcal{L}''_\phi(\phi_\star)$ converges to a cycle of period $k > 1$. Set $\eta = \eta_\phi/2$ and it is easy to convert the cycle back to the (w_1, w_2) space with the same period via the Lemma of equivalence. The dataset is non-separable by construction. □

Theorem (Existence of Cyclic Trajectory of 2-class Multinomial Logistic Regression in 2D)

For any $\gamma \in (0, 1]$, there exists a 2-dimensional non-separable 2-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size $\eta = \gamma/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period $k > 1$.

Proof Sketch. For any $\gamma \in (0, 1]$, Theorem 3 of Meng et al. (2024) constructs a 2D binary logistic regression dataset with stepsize $\eta_\phi = \gamma/\mathcal{L}''_\phi(\phi_\star)$ converging to a cycle. Set $\eta = \eta_\phi/2$ via the Lemma of equivalence again lifts the results back to the 2-class multinomial setting. □

Lemma (Higher-Dimensional Cyclic Trajectory Construction by Dataset Stacking)

Let $d', n' \geq 1, K \geq 2$, $\gamma \in (0, 1]$ and suppose there exists a dataset $(\tilde{\mathbf{X}}_{n' \times d'}, \tilde{\mathbf{y}}_{n' \times 1}) = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i \in [n']}$ with $\mathbf{x}^{(i)} \in \mathbb{R}^{d'}, y^{(i)} \in [K]$ such that gradient descent on the multinomial logistic loss with stepsize η_0 converges to a cycle of period $k > 1$. Then, for any integer $m \geq 1$, there exists a $d = md'$-dimensional dataset $(\mathbf{X}_{mn' \times md'}, \mathbf{y}_{mn' \times 1})$, where $\mathbf{X} = \mathbf{I}_m \otimes \tilde{\mathbf{X}}$ and $\mathbf{y} = (\tilde{\mathbf{y}}; \tilde{\mathbf{y}}; \ldots; \tilde{\mathbf{y}})$, such that gradient descent with stepsize $\eta = m\eta_0$ converges to a cycle of period k. Moreover, the eigenvalue of the Hessian matrix at the minimizer, of the new problem, is scaled down by m when compared to the original problem.

Proof Sketch. The new dataset consists of m independent blocks, each a copy of the original dataset embedded in orthogonal coordinates. The loss separates:

$$\mathcal{L}(\mathbf{W}) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}_i(\mathbf{W}_i).$$

The Hessian is block-diagonal with each block scaled by $1/m$; hence the largest eigenvalue becomes $\lambda_{\max}/m$.

Lemma (Higher-Dimensional Cyclic Trajectory Construction by Dataset Stacking)

Let $d', n' \geq 1, K \geq 2$, $\gamma \in (0, 1]$ and suppose there exists a dataset $(\tilde{\mathbf{X}}_{n' \times d'}, \tilde{\mathbf{y}}_{n' \times 1}) = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i \in [n']}$ with $\mathbf{x}^{(i)} \in \mathbb{R}^{d'}, y^{(i)} \in [K]$ such that gradient descent on the multinomial logistic loss with stepsize η_0 converges to a cycle of period $k > 1$. Then, for any integer $m \geq 1$, there exists a $d = md'$-dimensional dataset $(\mathbf{X}_{mn' \times md'}, \mathbf{y}_{mn' \times 1})$, where $\mathbf{X} = \mathbf{I}_m \otimes \tilde{\mathbf{X}}$ and $\mathbf{y} = (\tilde{\mathbf{y}}; \tilde{\mathbf{y}}; \ldots; \tilde{\mathbf{y}})$, such that gradient descent with stepsize $\eta = m\eta_0$ converges to a cycle of period k. Moreover, the eigenvalue of the Hessian matrix at the minimizer, of the new problem, is scaled down by m when compared to the original problem.

Proof Sketch (cont). The GD update for block i is

$$\mathbf{W}_i^{(t+1)} = \mathbf{W}_i^{(t)} - \eta \cdot \frac{1}{m} \nabla \mathcal{L}_i(\mathbf{W}_i^{(t)}) = \mathbf{W}_i^{(t)} - \eta_0 \nabla \mathcal{L}_i(\mathbf{W}_i^{(t)}),$$

where η_0 is the original stepsize. Since each block follows the original cycle, the full system cycles with the same period. □

Theorem (Existence of Cyclic Trajectory of 2-class Multinomial Logistic Regression in dD)

For any $\gamma \in (0, 1]$ and $d > 2$, there exists a d-dimensional non-separable 2-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size $\eta = \gamma/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period $k > 1$.

Proof Sketch. For any $\gamma \in (0, 1]$, take the 2D base dataset. Apply the stacking lemma with $m = \lfloor d/2 \rfloor$ and $d' = 2$ to obtain a dataset in dimension $2m \leq d$. The loss separates across blocks, the Hessian eigenvalues scale down by $1/m$, and the stepsize scales up by m. Consequently, the GD update for each block is identical to the original 2D cycle update. The full system therefore cycles with the same period k. If d is odd, we pad with an extra dimension containing only zero features (and set the corresponding weight component to zero); this dimension will not affect the cyclic trajectory and yields a d-dimensional non-separable 2-class problem with the desired cycle. □

Hypothesis

Hypothesis (Existence of Cyclic Trajectory of Multiclass Logistic Regression with More than 2 Classes)

For any $\gamma \in (0, 1]$ and any $d \geq 1, K \geq 3$, there exists a d-dimensional non-separable K-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size $\eta = \gamma/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period $k > 1$.

Directions:

1. Construct a family of datasets that induces cyclic behaviour by GD for any $K > 2$ in 1D $\Rightarrow$ Apply the stacking argument
2. Postulation: ≥ 3 classes gives enough freedom for rich dynamics, such as spiral-like oscillations or periodic orbits, supported by the rotational or reflective symmetries.

Orthogonal Weight Decoupling for K-Class

Theorem (Orthogonal Weight Decoupling for K-Class Cycle Construction)

For any integer $K \geq 2$ and any $\gamma \in (1, 2]$, there exists a non-separable K-class logistic regression problem of the form

$$\min_{\mathbf{w}_1,\ldots,\mathbf{w}_K \in \mathbb{R}^K} \mathcal{L}(\mathbf{w}_1, \ldots, \mathbf{w}_K) = -\frac{1}{N} \sum_{i=1}^{N} \log \left(\frac{\exp(\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x}^{(i)})} \right)$$

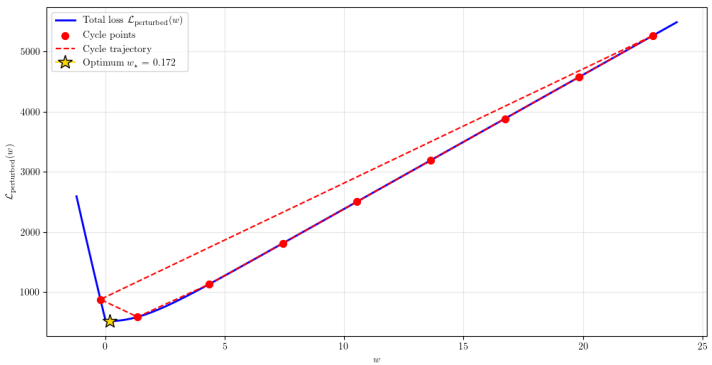
and a constant stepsize $\eta = \frac{\gamma}{\lambda_{\max}(\nabla^2 \mathcal{L}(\mathbf{w}_{1,\star},\ldots,\mathbf{w}_{K,\star}))}$, where $(\mathbf{w}_{1,\star}, \ldots, \mathbf{w}_{K,\star})$ is the unique minimizer of the problem, such that gradient descent with stepsize η converges to a stable cycle of period $k > 1$ for some initialization $\mathbf{w}_1, \ldots, \mathbf{w}_K$.

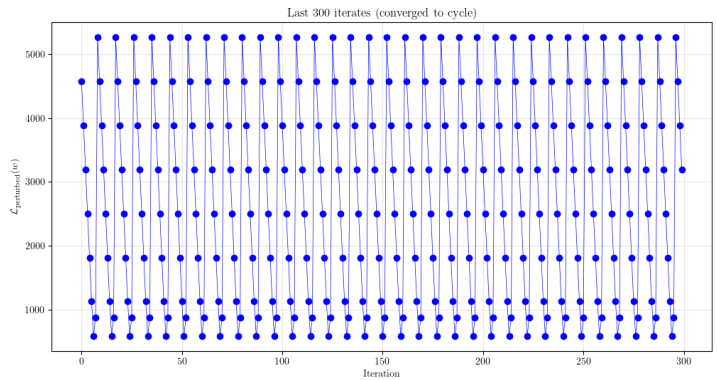
Idea. Place each class's feature on disjoint coordinates.

- For class k: include $m_k + n_k$ features of $\pm\mathbf{e}_k$ and extra large-magnitude samples to induce cycles when choosing the stepsize $\eta_k = \gamma/\lambda_k$.
- Initialize $\mathbf{w}_k = \theta_k \mathbf{e}_k$.
- Loss decouples into K independent 1D losses $\mathcal{L}_k(\theta_k)$ with period $\mathrm{lcm}(T_1, T_2, \ldots, T_k)$, apply scaling to achieve the same effective stepsize.

Cycle Construction for K-Class

Example (Cycle Construction for $\mathcal{L}_k$ with $K = 3$ and $\gamma = 1.5$)





Consider the dataset consisting of $n = 250$ copies of $x^{(i)} = 1$ and $m = 223$ copies of $x^{(i)} = -1$. On top of this dataset, we add 25 copies of $x^{(i)} = 65$ to get $\mathcal{L}_{\text{perturbed}}$. Start at $w_0 = 3.1$, and use $\gamma = 1.5$.
Particularly, for this case, we have $\mathcal{L}''_{\text{perturbed}}(w_\star) \approx 99.358596$. The yellow star marks the minimizer $w_\star \approx 0.333356$, the GD is run on $\mathcal{L}_{\text{perturbed}}$ with $\eta = \gamma / \mathcal{L}''_{\text{perturbed}}(w_\star) \approx 0.015097$, which converges to a period-9 cycle.

Conclusion

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- **Federated Learning with Heterogeneous Devices**
  1. `FedAvg` under device heterogeneity provably converges under multinomial logistic loss, for any stepsizes and any distribution of $\{T_i\}$;
  2. Once the stable phase is entered, last iterate convergence is $\mathcal{O}(1/(\bar{\eta}T_{\text{avg}}\gamma^2 R))$, implying that the heterogeneity effect vanishes asymptotically.
- **Non-Separable Multinomial Logistic Regression**
  1. Adaptive schedule $\eta_t = \alpha/(K\mathcal{L}(\mathbf{W}^{(t)})) \Rightarrow$ Monotonic convergence to a finite minimizer
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Large stepsizes can be **powerful**, **predictable**, and **principled**, with theoretical clarity and practical assurance.

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Thank you!